derive specific benefit from high-dose melphalan–ASCT and which patients do not need this treatment. Data from subgroup analyses in MIDAS suggest a benefit with high-dose melphalan–ASCT as compared with quadruplet therapy with regard to the occurrence of premaintenance MRD negativity at $10^{-6}$ sensitivity in patients with certain high-risk cytogenetic abnormalities, particularly the t(4;14) translocation[1]; these findings are consistent with differential progression-free survival benefits seen in the DETERMINATION trial.[3] Furthermore, data from MIDAS indicate a benefit of tandem high-dose melphalan–ASCT over single high-dose melphalan–ASCT in some MRD-positive patients with advanced-stage disease.[1]

Assessment of a patient's MRD status alone may not be sufficient for treatment decision making. However, MRD status combined with genomic risk stratification and other clinical features could provide a powerful approach to real-time tailoring of treatment that would enable us to determine how much treatment is truly enough to maximize benefit in our patients.

Disclosure forms provided by the authors are available with the full text of this editorial at NEJM.org.

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# Reducing the Risks of Mitochondrial Disease in Children

Robin Lovell-Badge, Ph.D.[1]

In vitro fertilization is clearly a successful technology, although it remains experimental; the safety and efficacy of new methods and reagents can be difficult to assess. And "traditional" reproduction is unsafe among carriers of pathogenic gene variants[1] because would-be parents may not know that they are carriers of genetic diseases, and de novo mutations can arise in each generation. In addition, the environment in which the embryo develops can accentuate epigenetic risks, which can affect the resulting child or adult.

How should ethical clinical research or trials involving early human embryos be conducted and how should appropriate regulation and oversight be established when the methods used could have an effect on the child? And what if the research involves deliberately making genetic alterations that can be heritable, with the aim of avoiding a genetic disease but also with possible, even if rare, unintended consequences, not just to the resulting child but to subsequent generations?

These questions, routinely asked in human embryo research, are especially relevant to two articles that describe a study led by Mary Herbert and Doug Turnbull, of Newcastle, United Kingdom, now reported in the *Journal*. Hyslop et al.[2]



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and McFarland et al.[3] report on the clinical use of pronuclear transfer — one method of mitochondrial replacement therapy or mitochondrial donation — as a way to avoid transmission of severe mitochondrial DNA (mtDNA) diseases. They began the study several years ago, after obtaining regulatory approval in 2015; the articles, which were hotly anticipated, show a (cautiously) good outcome and are well worth the wait.

There are several relevant examples of clinical research involving new methods associated with in vitro fertilization — some helpful, others not. Preimplantation genetic testing (PGT) and intracytoplasmic sperm injection, both relevant to the study reported by Hyslop et al. and McFarland et al., have contrasting histories. PGT was introduced after preclinical research in mice and human embryos,[4] whereas the first successful attempts at intracytoplasmic sperm injection in human in vitro fertilization occurred after an accident in which the glass needle being used to introduce a single sperm under the zona pellucida (an established method used in cases of very low sperm count) penetrated the membrane of the oocyte as the sperm was being expelled.[5,6] Another example, and one that should not be emulated, is the attempt at heritable genome editing by He Jiankui.[7] This attempt was replete with many problems, including the lack of a good plan for follow-up of the three resulting girls; after 7 years, there is no available independent assessment of the nature of any genetic alteration or the health status of the girls. Follow-up is particularly critical to inform practice but must be done with care to avoid over-medicalizing the child and subjecting parents to unwanted levels of anxiety.

The biology of mitochondria and the causes of mtDNA diseases and the ways to avoid them are complex topics,[2,3,8] as explained by Julie Steffann in an accompanying editorial in the *Journal*.[9] This complexity, concern about heritability, and some strong opposition contributed to a very long pathway leading to a change in U.K. regulations that permitted the work of the Newcastle group (Fig. 1). However, what was pressingly clear is that the disorders associated with deleterious variants of mtDNA can be severe and can put a heavy burden on parents and health care systems. Although adoption and egg or embryo donation are possible solutions, they do not allow mothers to have their own genetically related child, which is often of paramount importance to parents. Given that transmission of mtDNA disease is not always certain (although it can be), some women will try again and again, each time having a child who may suffer and die.

The notion of mitochondrial replacement as a solution has been around for 30 years. If we set aside the possibility of genome editing of mtDNA, which is currently too risky, three main methods could be used: pronuclear transfer (carried out after fertilization), maternal-spindle transfer (carried out between unfertilized oocytes), and polar-body transfer. This latter method is the newest option, with relatively little preclinical data, but a recent report[20] implies that it has been used successfully in one case. Pronuclear transfer is a relatively established method, with substantial preclinical research, and the one on which the Newcastle team has focused.

Zhang et al.[21] reported using maternal-spindle transfer to prevent transmission of the Leigh syndrome (a type of mitochondrial disease): they derived an embryo in New York and then shipped it to Mexico for implantation into the mother in an attempt to avoid breaking laws in either country. The publicity around this case prevented follow-up assessments, so we do not know whether the child remains healthy, or whether there is any indication that the methods used needed improvement. Maternal-spindle transfer has also been used to treat idiopathic infertility, a practice that is not permitted in the United Kingdom. Despite some evidence of success,[22] how this procedure might work is not understood.

The Newcastle team opted to compare pronuclear transfer with PGT,[2] the latter method being a standard way to reduce pathogenic mtDNA load, with substantial evidence that it can lead to children being born without disease. However, the approach involving PGT requires that at least one derived embryo has a sufficiently low level of heteroplasmy (in which mtDNA is a mix of two or more sequences) to justify intrauterine transplantation (i.e., derived progeny would have a substantially reduced risk of disease), an impossible outcome for women who have homoplasmy (in which all copies of mtDNA have the same sequence) or who have very high levels of heteroplasmy. The Newcastle group therefore gave such



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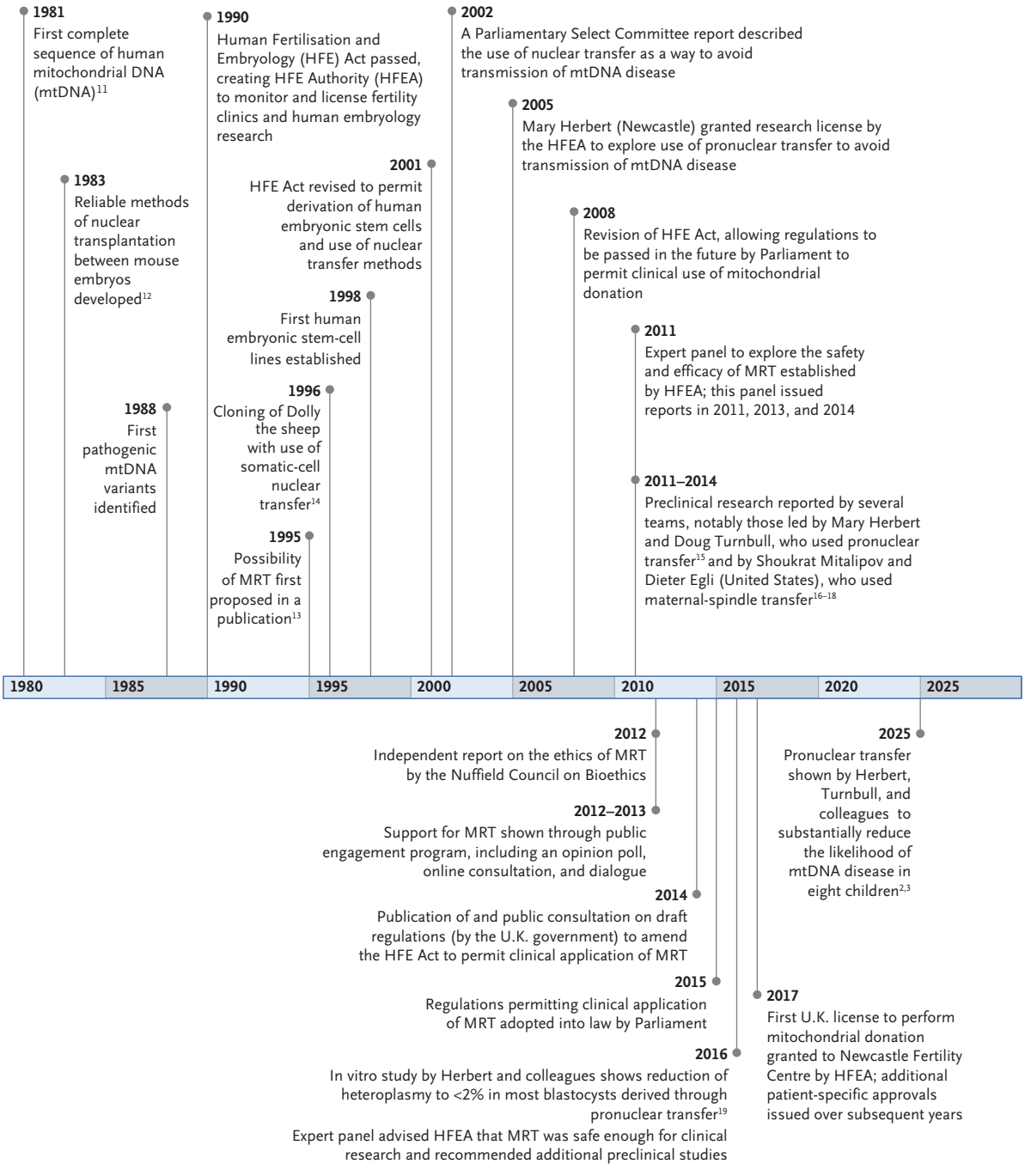



**Figure 1. Mitochondrial Replacement Therapy (MRT) in the United Kingdom.**

Shown is a timeline of events beginning in 1981, including a change in U.K. regulations to allow clinical research on MRT, and ending with reports of eight children born after pronuclear transfer.[2,3,10-19]



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women the option of pronuclear transfer. Thus, the group of patients who underwent pronuclear transfer and the group that underwent PGT were different in various ways, including not being matched with respect to specific diseases and mtDNA variants. However, having the group that underwent PGT as the comparison group strengthens the conclusion that pronuclear transfer is a valid method to reduce the risk of mtDNA disease. Many of the outcomes assessed were nearly equivalent in the study groups, including similar percentages of patients who had a pregnancy, although the percentage of patients who had embryo loss was higher after pronuclear transfer than after PGT, for reasons that are unclear. A total of 18 live births occurred after PGT, each with a sufficiently low proportion of the pathogenic mtDNA to confirm that this remains a valid risk-reduction method. The fact that eight apparently healthy babies were born after pronuclear transfer (without which mitochondrial disease would have developed in these babies) is a clear demonstration of the usefulness of this method among women for whom PGT is not an option.

The Newcastle group has not only implemented mitochondrial donation as an option for women with mitochondrial disease. They also designed and implemented an entire program of clinical care to address mitochondrial disease in the context of reproduction,[3] where patients are offered options that minimize risks and provide the best likelihood for them to have a healthy child. The long road to this point has no doubt been frustrating to women at risk for having children with mtDNA disease (and to their partners and families) and to scientists and clinicians involved in finding solutions. However, this cautious approach has been necessary to ensure that the public, policymakers, scientists, and health care professionals are aligned with respect to the use of a ground-breaking method.

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